

Reference Model-Based Cyber-Attack Detection for Stochastically Uncertain Wind Turbine Systems

Álvaro Martín Gómez[✉], Sina Hassani[✉], Szymon Greś[✉], Rafał Wiśniewski[✉], and Steven X. Ding[✉]

Abstract—As wind turbines become integral to critical infrastructure, ensuring their resilience against faults and cyber-attacks is essential. This work implements a reference model-based anomaly detection scheme for wind turbines with stochastic parametric uncertainty. An $\mathcal{H}_i/\mathcal{H}_\infty$ reference filter balances fault sensitivity and disturbance rejection, while a robust filter is designed via an $\mathcal{H}_\infty$ model matching problem to track reference residuals while bounding the effect of uncertainty. This bound is made less conservative by weighting residual components based on their sensitivity to uncertainty. Residual covariance due to parametric uncertainty is derived from a statistical analysis of the residual generator dynamics and applied in a Generalized Likelihood Ratio test. This provides a rigorous alternative to heuristic covariance inflation by capturing the directional properties of uncertainty in the residual subspace. Experimental results compare the residuals and uncertainty bounds of the reference and robust filters. The proposed method reduces residual sensitivity to changes in operating conditions as well as decision function variability due to parametric uncertainty.

Index Terms—wind energy, anomaly detection, robust filters, stochastic uncertainties.

I. INTRODUCTION

CYBER threats to the energy sector, particularly wind generation, are escalating due to sophisticated attacks by nation-state actors and financially motivated criminals. Common attack vectors include ransomware (e.g., Nordex, Deutsche Windtechnik), supply chain compromises, and exploitation of third-party access, often leading to operational disruptions, loss of remote monitoring, or reputational damage [1]. For instance, the 2022 Viasat satellite modem attack disabled communication with 5,800 wind turbines [2], while ransomware incidents forced temporary disconnections from control systems. Unlike overt attacks like ransomware, where detection is trivial but prevention plays a crucial role, false data injection attacks (FDIA) may evade traditional safeguards, and pose a subtler risk by manipulating operational data to destabilize grid decisions without immediate detection. Hence, they are the focus of this work.

The data-driven methods presented in [3]–[5], among others, provide effective approaches to anomaly detection without

the need for complex models or control theory concepts. For instance, adaptive intrusion detection frameworks combining machine learning with residual-based anomaly detection have been proposed to identify false data injection attacks in smart power systems [6]. However, the lack of underlying structure in black-box models can make long-term predictions unreliable in unsupervised settings, while limited access to datasets, especially for cyber-attack scenarios, reduces the practicality and effectiveness of supervised learning methods in real-world applications.

Conversely, the main challenge of model-based detection lies in the availability of an accurate model for the system under consideration [7]. Regarding wind turbines, finite element models, e.g. those based on Blade Element Momentum (BEM) theory [8], are useful for simulation but too computationally intensive for real-time FDI. Simplified models from one-dimensional (1-D) momentum theory [8] are more practical but less accurate, making it harder to separate true faults from modelling errors. This challenge is amplified when proprietary aerodynamic data, such as power and thrust coefficient tables, are unavailable, which is often the case for non-manufacturing companies.

Recent contributions on wind turbine fault diagnosis have explored zonotopic observer-based approaches combined with neuro-fuzzy or TS/MANFIS modelling, enabling set-membership state estimation and robust fault detection under bounded uncertainties [9], [10]. These methods provide guaranteed consistency bounds and have shown strong performance for fault detection under set-bounded uncertainty descriptions. In this work, a different perspective is adopted by focusing on stochastic parametric uncertainty, with the aim of enabling a probabilistic characterization of residuals that can be directly used in statistical change detection.

The field of fault detection filter (FDF) design has evolved significantly through various methodological approaches aimed at achieving robust fault detection. Early techniques focused on developing FDFs that generate residuals sensitive to faults by rejecting unknown inputs [11]–[13]. However, such complete decoupling often comes at the cost of reduced sensitivity to actual faults, limiting detection performance [14]. Other approaches have employed $\mathcal{H}_\infty/\mathcal{H}_\infty$ optimization frameworks to simultaneously minimize disturbance effects while maximizing the maximum influence of faults [15], as well as $\mathcal{H}_-/\mathcal{H}_\infty$ formulations that specifically maximize the minimum singular value of the transfer matrix from faults to residuals [16], [17]. The $\mathcal{H}_i/\mathcal{H}_\infty$ "unified

Á. M. Gómez, S. Hassani, S. Greś, and R. Wiśniewski are with the Department of Electronic Systems, Aalborg University, Aalborg, Denmark (e-mail: almago@es.aau.dk; sinah@es.aau.dk; sgres@es.aau.dk; raf@es.aau.dk).

S. X. Ding is with the Department of Automatic Control and Complex Systems, University of Duisburg–Essen, Duisburg, Germany (e-mail: steven.ding@uni-due.de).

This work was supported by the BeWind project, funded by Innovation Fund Denmark.

This paper was presented in part at the IAS Annual Meeting, July 9, 2025. DOI: 10.1109/IAS62731.2025.11061699

solution” generalizes both approaches for linear time-invariant [18], [19] and time-varying systems [20], providing an optimal compromise between disturbance robustness and fault sensitivity through algebraic solutions that guarantee performance across the entire residual subspace and frequency range [21].

However, while these methods effectively handle unknown inputs and stochastic noise, they remain fundamentally limited in addressing parametric uncertainties that affect system dynamics and sensor/actuator parameters. Such parameter variations corrupt residual signals, leading to performance degradation through increased false alarms and missed detections. The reference model-based approach [14], [22]–[24] addresses these challenges by formulating fault detection as an $\mathcal{H}_\infty$ model-matching problem (MMP). The method uses a reference model, which serves as an optimal residual generator under the assumption of no model uncertainties aside from disturbances. Simultaneously, it aims to minimize the $\mathcal{H}_\infty$ norm of the difference between the designed residual generator and this reference model. With model parameters estimated via Bayesian inference, modelling errors are treated as stochastic uncertainties, leading to an LMI solution originally derived in Chapter 8 in [21].

This anomaly detection scheme has been given very little attention, mainly due to its complexity and its tendency to become overly conservative. To the best of the authors’ knowledge, [25] is the only related application, which specifically focuses on an open-loop wind turbine pitch actuator and drive-train system for sensor and actuator fault detection. However, the paper omits critical details regarding the uncertainties chosen in the system parameters and lacks an evaluation of the residual’s robustness improvements relative to the standard $\mathcal{H}_i/\mathcal{H}_\infty$ unified solution when these uncertainties are present. These gaps highlight the need for a more thorough and well-documented application of this technique to key wind turbine controllers, with a clearer demonstration of its advantages and limitations and adjustments to some of the metrics to reduce conservativeness. This paper is an extended version of the work presented at the 2025 IEEE Industry Applications Society Annual Meeting [26].

II. STRATEGY OVERVIEW AND ATTACK SCENARIOS

The anomaly detection scheme of this work adopts a reference model-based strategy, illustrated in Fig. 1, to address parametric uncertainty while preserving fault sensitivity. The approach separates performance specification from robust implementation through a two-step design. First, a reference residual generator is constructed using the nominal model. In this work, the $\mathcal{H}_i/\mathcal{H}_\infty$ “unified solution” is selected as reference residual generator, as it provides an optimal trade-off between disturbance attenuation and fault sensitivity across the residual space [7], [21]. This reference defines the desired residual behaviour. Second, a robust residual generator is obtained by solving an $\mathcal{H}_\infty$ model-matching problem (MMP) [21], where the objective is to minimize the deviation between the implemented residual and the reference residual in the presence of uncertainties. This formulation ensures that the residual closely reproduces the nominal optimal response

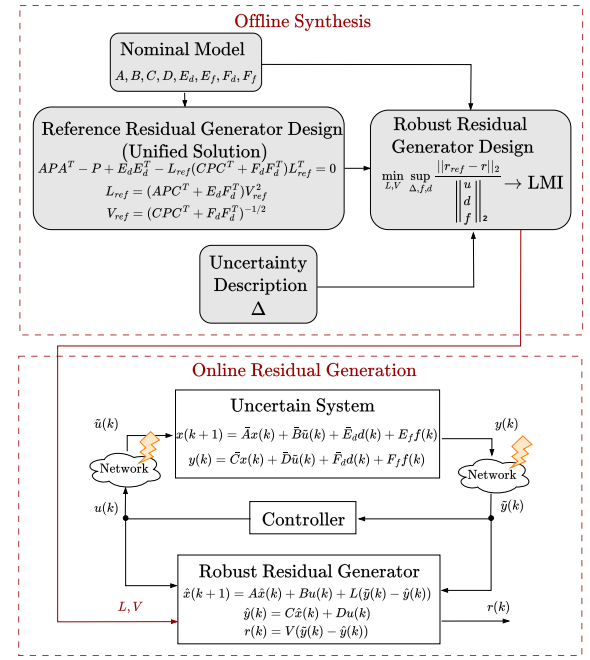


Fig. 1. $\mathcal{H}_\infty$ MMP using $\mathcal{H}_i/\mathcal{H}_\infty$ FDF as reference model for anomaly detection in uncertain systems.

while explicitly accounting for disturbances and parametric variations. As shown in Fig. 1, the design is performed offline, and the resulting robust filter is used online for residual generation and subsequent anomaly detection.

Following the structure proposed in [27], the filter receives potentially corrupted sensor measurements together with control signals that remain unaltered at the point of filtering, since any manipulation of these signals would occur downstream in the communication network. However, these uncorrupted control signals may still reflect the effects of attacks due to the feedback loop.

In this work, we focus on false data injection attacks (FDIA), i.e., integrity attacks that directly manipulate sensor and actuator signals in the control loop. In contrast to availability attacks such as denial-of-service, which are typically straightforward to detect due to missing or blocked data, FDI attacks alter critical variables while preserving normal communication patterns, making them significantly more challenging to detect and potentially more damaging for closed-loop operation.

The chosen attack scenario corresponds to a bias injection attack, a subclass of FDIA introduced in [27]. This attack consists of injecting a slowly varying bias into selected signals, with the objective of gradually driving the system away from its nominal operating condition and potentially into an unsafe region. This class of attacks requires only limited model knowledge and no access to the full system state, making it representative of realistic adversaries with constrained resources.

The attack is generated through a first-order low-pass dynamics, allowing the adversary to regulate the rate of change of the injected signal via the gain parameter a_∞ and time constant β_a , as described in (1) and shown in Fig. 2. This structure

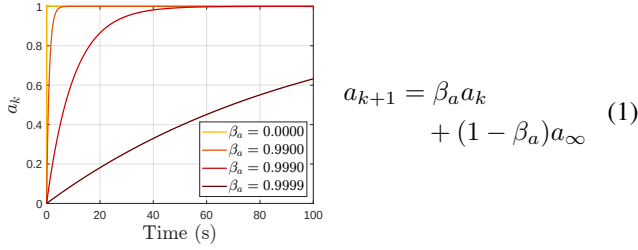


Fig. 2. Bias injection attack for different β_a , with a sampling time of 0.01 s.

produces deviations that evolve gradually over time, which are known to be difficult to distinguish from model uncertainty and disturbances in model-based detection schemes. Rather than optimizing the stealth level as in [27], the attack parameters are selected to enable a clear comparison between the standard and robust residual filters.

III. METHODS

A. Unified Solution ($\mathcal{H}_i/\mathcal{H}_\infty$) Reference Filter

Consider the discrete-time system

$$x(k+1) = Ax(k) + Bu(k) + E_d d(k) + E_f f(k), \quad (2)$$

$$y(k) = Cx(k) + Du(k) + F_d d(k) + F_f f(k), \quad (3)$$

where $x(k) \in \mathbb{R}^n$, $u(k) \in \mathbb{R}^p$, and $y(k) \in \mathbb{R}^m$ denote the state, input, and measured output, respectively, while $d(k) \in \mathbb{R}^{k_d}$ and $f(k) \in \mathbb{R}^{k_f}$ represent unknown disturbances and faults. The objective is to construct a residual signal $r(k)$ that is sensitive to faults while robust to disturbances, leading to a fundamental trade-off between these two requirements [21].

Under the assumptions

$$\text{rank}(G_{yf}(z)) = m, \quad G_{yf}(z) = C(zI - A)^{-1}E_f + F_f, \quad (4)$$

$$\forall \theta \in [0, 2\pi], \quad \text{rank} \begin{bmatrix} A - e^{j\theta}I & E_d \\ C & F_d \end{bmatrix} = n + m, \quad (5)$$

the residual generator is constructed using an observer-based structure

$$\hat{x}(k+1) = A\hat{x}(k) + Bu(k) + L(y(k) - \hat{y}(k)), \quad (6)$$

$$\hat{y}(k) = C\hat{x}(k) + Du(k), \quad (7)$$

$$r(k) = V(y(k) - \hat{y}(k)). \quad (8)$$

The gains L and V are selected to optimize the trade-off between disturbance attenuation and fault sensitivity. The resulting optimal solution is given by

$$L_{\text{opt}} = (APC^T + E_d F_d^T) V_{\text{opt}}^2, \quad (9)$$

$$V_{\text{opt}} = (CPC^T + F_d F_d^T)^{-1/2}, \quad (10)$$

where $P > 0$ is the stabilizing solution of the Riccati equation

$$APA^T - P + E_d E_d^T - L_{\text{opt}}(CPC^T + F_d F_d^T)L_{\text{opt}} = 0. \quad (11)$$

The optimality of this solution is characterized by the $\mathcal{H}_i/\mathcal{H}_\infty$ performance index

$$\{L_{\text{opt}}, V_{\text{opt}}\} = \arg \sup_{L, V} \frac{\sigma_i(V \hat{N}_f(e^{j\theta}))}{\|V \hat{N}_d\|_\infty}, \quad \forall \theta \in [0, 2\pi], \quad (12)$$

with

$$\hat{N}_f(z) = F_f + C(zI - A + LC)^{-1}(E_f - LF_f), \quad (13)$$

$$\hat{N}_d(z) = F_d + C(zI - A + LC)^{-1}(E_d - LF_d), \quad (14)$$

describing the fault and disturbance propagation to the residual signal. In particular, when $d(k)$ is modelled as white noise with identity covariance, the resulting residual generator reduces to the steady-state Kalman filter.

B. Robustness Against Stochastic Uncertainties via an $\mathcal{H}_\infty$ Model-Matching Problem

Consider the system defined by

$$x(k+1) = \bar{A}x(k) + \bar{B}u(k) + \bar{E}_d d(k) + E_f f(k), \quad (15)$$

$$y(k) = \bar{C}x(k) + \bar{D}u(k) + \bar{F}_d d(k) + F_f f(k), \quad (16)$$

$$\bar{A} = A + \Delta A, \quad \bar{B} = B + \Delta B, \quad \bar{C} = C + \Delta C,$$

$$\bar{D} = D + \Delta D, \quad \bar{E}_d = E_d + \Delta E, \quad \bar{F}_d = F_d + \Delta F,$$

where the uncertainty terms ΔA , ΔB , ΔC , ΔD , ΔE , and ΔF satisfy

$$\begin{bmatrix} \Delta A & \Delta B & \Delta E \\ \Delta C & \Delta D & \Delta F \end{bmatrix} = \sum_{i=1}^l \begin{bmatrix} A_i & B_i & E_i \\ C_i & D_i & F_i \end{bmatrix} p_i(k), \quad (17)$$

where A_i , B_i , C_i , D_i , E_i , F_i , $i = 1, \dots, l$, are known matrices of appropriate dimensions. The vector $p(k) = [p_1(k) \dots p_l(k)]^T$ is a stochastic process with

$$\mathbb{E}(p(k)) = 0, \quad \mathbb{E}(p(k)p^T(k)) = \text{diag}(\sigma_1, \dots, \sigma_l),$$

where $\sigma_i > 0$, $i = 1, \dots, l$, are known. It is further assumed that $p(0)$, $p(1)$, $\dots$, are independent and $x(0)$, $u(k)$, $d(k)$, $f(k)$ are independent of $p(k)$.

The residual generator is given by

$$\hat{x}(k+1) = A\hat{x}(k) + Bu(k) + L(y(k) - \hat{y}(k)), \quad (18)$$

$$\hat{y}(k) = C\hat{x}(k) + Du(k), \quad (19)$$

$$r(k) = V(y(k) - \hat{y}(k)), \quad (20)$$

where (L, V) are to be designed.

The objective is to determine L and V such that the residual $r(k)$ follows the reference residual $r_{\text{ref}}(k)$, obtained from the unified solution, while accounting for stochastic uncertainties. This leads to the optimization problem

$$\min_{L, V} \mathbb{E} \left[(r(k) - r_{\text{ref}}(k))^T (r(k) - r_{\text{ref}}(k)) \right], \quad (21)$$

which captures the model-matching objective in a stochastic setting.

Using the decomposition

$$\begin{aligned} & \mathbb{E} \left[(r(k) - r_{\text{ref}}(k))^T (r(k) - r_{\text{ref}}(k)) \right] \\ &= (\bar{r}(k) - r_{\text{ref}}(k))^T (\bar{r}(k) - r_{\text{ref}}(k)) + \sigma_r^2(k), \end{aligned} \quad (22)$$

with $\bar{r}(k) = \mathbb{E}[r(k)]$, the problem can be reformulated as minimizing the deviation of the mean residual from the reference model, while bounding the residual variance

$$\sigma_r^2(k) = \mathbb{E} \left[(r(k) - \bar{r}(k))^T Q_r (r(k) - \bar{r}(k)) \right] \leq \delta, \quad (23)$$

for some $\delta > 0$, where $Q_r \succeq 0$ is a weighting matrix. Compared to the standard formulation, the introduction of Q_r allows selective penalization of residual components that are more sensitive to parametric uncertainty. This enables reducing conservativeness by relaxing the variance constraint in less critical directions, while preserving robustness in the most relevant residual channels for fault detection.

To derive tractable conditions, the design problem is expressed in terms of the following inequalities:

$$\begin{aligned} & (\bar{r}(k) - r_{\text{ref}}(k))^T (\bar{r}(k) - r_{\text{ref}}(k)) \\ & < \alpha_1^2 \sum_{j=0}^{k-1} (d^T(j)d(j) + f^T(j)f(j)) \\ & + \alpha_2^2 (d^T(k)d(k) + f^T(k)f(k)), \end{aligned} \quad (24)$$

and

$$\begin{aligned} \sigma_r^2(k) & < \gamma_1^2 \sum_{j=0}^{k-1} (d^T(j)d(j) + f^T(j)f(j) + u^T(j)u(j)) \\ & + \gamma_2^2 (d^T(k)d(k) + f^T(k)f(k) + u^T(k)u(k)), \end{aligned} \quad (25)$$

for some $\alpha_1 > 0$, $\alpha_2 > 0$, $\gamma_1 > 0$, and $\gamma_2 > 0$.

An LMI-based solution can be derived using the Bounded Real Lemma. For the full derivation and corresponding LMI conditions, see [21]. Appendix A provides the reformulated LMIs associated with (25) incorporating the weighted residual variance.

C. Statistical Change Detection

The residual signal $r(k)$ is evaluated using a statistical change detection scheme aimed at identifying deviations from its nominal behaviour. In the absence of faults or attacks, the residual is expected to fluctuate around a nominal mean μ_0 . The occurrence of faults or malicious manipulations induces a shift in this mean, which can be detected using hypothesis testing.

Over a sliding window of length M , define the sample mean $\bar{r}(k) = \frac{1}{M} \sum_{i=k-M+1}^k r(i)$. The residual sequence is assumed to be a Gaussian process with constant covariance Σ_r , and samples within the sliding window are assumed to be independent and identically distributed. Furthermore, the pre-change mean μ_0 and covariance Σ_r are assumed known, while the post-change mean μ_1 is unknown. Under these assumptions, a mean-shift detection problem is formulated as

$$\begin{aligned} \mathcal{H}_0 : & \quad r(k) \sim \mathcal{N}(\mu_0, \Sigma_r), \\ \mathcal{H}_1 : & \quad r(k) \sim \mathcal{N}(\mu_1, \Sigma_r), \quad \mu_1 \neq \mu_0, \end{aligned}$$

The generalized likelihood ratio (GLR) test for unknown post-change mean μ_1 leads to the decision statistic [28]

$$J(k) = \frac{M}{2} (\bar{r}(k) - \mu_0)^T \Sigma_r^+ (\bar{r}(k) - \mu_0), \quad (26)$$

where Σ_r^+ denotes the Moore–Penrose pseudoinverse (to cover singular Σ_r). Let $m = \text{rank}(\Sigma_r)$ be the effective residual dimension. Under $\mathcal{H}_0$, the statistic satisfies $2J \sim \chi_m^2$, while under $\mathcal{H}_1$, $2J \sim \chi_m^2(\lambda)$ with noncentrality parameter $\lambda = M(\mu_1 - \mu_0)^T \Sigma_r^+ (\mu_1 - \mu_0)$. Given a desired false-alarm probability α , the detection threshold γ is selected

from the $(1 - \alpha)$ quantile of the χ_m^2 distribution such that $\Pr(2J > \gamma \mid \mathcal{H}_0) = \alpha$.

However, due to the multiplicative interaction between uncertain parameters and states, as well as linearization errors and unmodelled dynamics, the residual sequence is in general temporally correlated and not strictly Gaussian. Consequently, the χ^2 distribution of the test statistic under $\mathcal{H}_0$ should be interpreted as an approximation. In practice, the resulting threshold provides a useful design guideline, although the actual false-alarm probability may deviate from the nominal value and empirical tuning may be required.

The performance of the GLR test depends critically on an accurate characterization of the residual covariance Σ_r . In this work, Σ_r is derived explicitly from the stochastic uncertainty model introduced in (15)–(16), ensuring consistency between filter design and statistical evaluation.

The residual covariance is defined as $\Sigma_r(k) = \mathbb{E}[(r(k) - \bar{r}(k))(r(k) - \bar{r}(k))^T]$, where $\bar{r}(k) = \mathbb{E}[r(k)]$. Using the residual dynamics derived in Appendix A, the deviation from the mean can be expressed as

$$\begin{aligned} r(k) - \bar{r}(k) &= C_{r,0} e_{x_r}(k) + \sum_{i=1}^l C_{r,i} p_i(k) x_r(k) \\ &+ \sum_{i=1}^l D_{r,i} p_i(k) u(k) + \sum_{i=1}^l F_{r,i} p_i(k) d(k), \end{aligned} \quad (27)$$

where $e_{x_r}(k) = x_r(k) - \bar{x}_r(k)$ is the augmented state error.

Taking expectations and exploiting the independence and second-order properties of $p(k)$ yields

$$\begin{aligned} \Sigma_r(k) &= C_{r,0} E_{x_r}(k) C_{r,0}^T \\ &+ \sum_{i=1}^l \sigma_i^2 C_{r,i} (\bar{x}_r(k) \bar{x}_r(k)^T + E_{x_r}(k)) C_{r,i}^T \\ &+ \sum_{i=1}^l \sigma_i^2 [D_{r,i} u(k) u(k)^T D_{r,i}^T + F_{r,i} d(k) d(k)^T F_{r,i}^T] \\ &+ \sum_{i=1}^l \sigma_i^2 [C_{r,i} \bar{x}_r(k) u(k)^T D_{r,i}^T + D_{r,i} u(k) \bar{x}_r(k)^T C_{r,i}^T] \\ &+ \sum_{i=1}^l \sigma_i^2 [C_{r,i} \bar{x}_r(k) d(k)^T F_{r,i}^T + F_{r,i} d(k) \bar{x}_r(k)^T C_{r,i}^T] \\ &+ \sum_{i=1}^l \sigma_i^2 [D_{r,i} u(k) d(k)^T F_{r,i}^T + F_{r,i} d(k) u(k)^T D_{r,i}^T], \end{aligned} \quad (28)$$

where $E_{x_r}(k) = \mathbb{E}[e_{x_r}(k) e_{x_r}(k)^T]$ is given in Appendix A.

This expression captures the contribution of parametric uncertainty, inputs, and disturbances to the residual variability. It can be evaluated for both the reference filter and the robust filter by substituting the corresponding gains (L, V) . The resulting covariance matrix Σ_r is then used in (26) to weight the residual components.

IV. MODEL AND PARAMETER ESTIMATION

This paper focuses on the aerodynamic/mechanical subsystem rather than the electrical subsystem [29], as its strong nonlinearities and complex behaviour make it a more compelling

subject of study. The aerodynamic behaviour of a wind turbine is derived using one-dimensional momentum theory for an ideal rotor, assuming a frictionless actuator disk with no rotational velocity component in the wake. This approach yields expressions for the extracted power and thrust force [8]

$$P = \frac{1}{2} \rho A v_r^3 C_p(\lambda, \beta), \quad (29)$$

$$F = \frac{1}{2} \rho A v_r^2 C_t(\lambda, \beta), \quad (30)$$

where $C_p(\lambda, \beta)$ and $C_t(\lambda, \beta)$ represent dimensionless power and thrust coefficients, A is the rotor swept area, ρ is air density, and v_r is the effective wind speed at the rotor plane. These coefficients are typically tabulated as functions of tip-speed ratio $\lambda = \omega_r R / v_r$ (where ω_r is rotor speed and R is rotor radius) and blade pitch angle β .

When proprietary look-up tables for the power coefficient are unavailable, a parametric approximation for $C_p(\lambda, \beta)$ [30] can be used:

$$C_p(\lambda, \beta) = c_1(c_2 \lambda_i^{-1} - c_3 \beta - c_4) e^{-c_5 \lambda_i^{-1}} + c_6 \lambda, \quad (31)$$

$$\lambda_i^{-1} = \frac{1}{\lambda + 0.08\beta} - \frac{0.035}{\beta^3 + 1}. \quad (32)$$

The aerodynamic torque follows from $P = T_r \omega$, yielding

$$T_r = \frac{1}{2} \rho A v_r^3 C_p(\lambda, \beta) / \omega_r, \quad (33)$$

where T_r denotes the rotor aerodynamic torque. This formulation assumes stationary flow conditions given by ω_r , β and v_r . Dynamic inflow effects [31] are neglected in this work due to their relatively small impact compared to parametric model uncertainties. This model is intended for normal above-cut-in operating regime (Region 2–3), where ω_r is bounded away from zero.

The wind turbine being modelled is equipped with a permanent magnet synchronous generator, which is used in conjunction with direct drive design. Direct drive turbines lack gearbox, so the flexibility of the drive train is considered negligible and thus, the generator speed ω_g is equal to the rotor speed ω_r . This rigid model is then described as

$$(J_r + J_g) \dot{\omega}_r = (1 - \mu_d) T_r - T_g, \quad (34)$$

where the parameter μ_d accounts for the mechanical losses, and J_r and J_g refer to the moments of inertia of the rotor and generator respectively.

Actuator dynamics are modelled as first-order lags of time constants τ_β and τ_g respectively:

$$\dot{\beta} = (\beta_{\text{ref}} - \beta) / \tau_\beta, \quad (35)$$

$$\dot{T}_g = (T_{g,\text{ref}} - T_g) / \tau_g. \quad (36)$$

Control strategies differ by operational region. For below-rated wind speed conditions, the pitch is normally set to 0° and the generator torque is set to its optimal value, which is computed as [32]

$$T_{g,\text{ref}} = K_{\text{opt}} \omega_r^2, \quad K_{\text{opt}} = \frac{1}{2} \frac{\rho R^3 A C_{p,\text{max}}}{\lambda_{\text{opt}}^3}, \quad (37)$$

where $C_{p,\text{max}}$ and λ_{opt} can be obtained from the wind turbine C_p/λ curve for $\theta = 0$ [8].

For above-rated wind speed conditions, a basic PI controller is used to compute the pitch reference based on the rotor speed deviation from its rated value ($\omega_{r,\text{err}}$), while the torque is regulated around a rated value $T_{g,\text{rated}}$ using a proportional gain to aid the pitch control in its task [32]:

$$\beta_{\text{ref}} = -K_{p,\beta} \dot{\omega}_{r,\text{err}} + K_{i,\beta} \omega_{r,\text{err}}, \quad (38)$$

$$T_{g,\text{ref}} = K_{p,T_g} \omega_{r,\text{err}} + T_{g,\text{rated}}. \quad (39)$$

Structural dynamics are omitted from this simplified model, as they do not significantly influence the behaviour of collective pitch and torque controllers within the conventional wind turbine control framework. Although the wind speed at the rotor plane is typically estimated using Kalman filtering based on tower-top accelerometer data and a wind model [33], this work treats wind speed as an exogenous input, assuming prior estimation in order to avoid additional model complexity.

The model (31)–(36) is linearized via a Taylor expansion around operating points defined by the effective wind speed at the rotor v_r , and discretized using forward Euler with sampling time of $T_s = 0.01$ s, sufficiently small relative to the dominant system dynamics to preserve stability. It consists of the following states and outputs, $y = x = [\omega_r \ T_g \ \beta]^T$ (full-state measurement) and exogenous/control inputs, $u = [v_r \ T_{g,\text{ref}} \ \beta_{\text{ref}}]^T$. In this work, the process and measurement disturbance distribution matrices in (2)–(3) are set as $E_d = 0$ and $F_d = I$, respectively, i.e., $d(k)$ is modelled as measurement noise only. Assumption (4) holds since F_f contains C , and thus $G_{yf}(z) = C(zI - A)^{-1} E_f + F_f$ has full row rank m for all z when C is full row rank. Moreover, with $E_d = 0$ and $F_d = I$, Assumption (5) reduces to requiring that A has no eigenvalues on the unit circle, which is satisfied. Under standard detectability/stabilizability assumptions, (11) admits a unique stabilizing solution $P \succeq 0$. With $F_d = I$, $R = F_d F_d^T = I \succ 0$, hence $CPC^T + R \succ 0$ is invertible and V_{opt} in (10) is well-defined.

In this study, three attack scenarios are considered, targeting the measurement of the rotor speed ω_r and the reference control signals $T_{g,\text{ref}}$ and β_{ref} . Measurements of T_g and β are not targeted, since they are not used directly by the standard controller considered in this work and altering them would have no practical effect. The three scenarios are defined as follows:

- *Test 1 (rotor speed measurement attack)*: bias injection in the measurement of ω_r with magnitude $a_\infty = 1 \cdot \omega_r^{\text{op}}$,
- *Test 2 (torque reference attack)*: bias injection in $T_{g,\text{ref}}$ with magnitude $a_\infty = -0.1 \cdot T_{g,\text{ref}}^{\text{op}}$,
- *Test 3 (pitch reference attack)*: bias injection in β_{ref} with magnitude $a_\infty = 0.1 \cdot \beta^{\text{op}}$.

A relatively high gain is used for ω_r to overcome the large estimation uncertainty associated with this variable, ensuring a distinguishable residual response. The time constant is fixed at $\beta_a = 0.999$ in (1), with a sampling interval of 0.01 s, resulting in a sufficiently abrupt injection profile for visualization purposes.

The constants used for the 15 MW reference wind turbine are listed in Table I, and a high-fidelity simulation of this turbine is available through GH Bladed software [34]. Table II

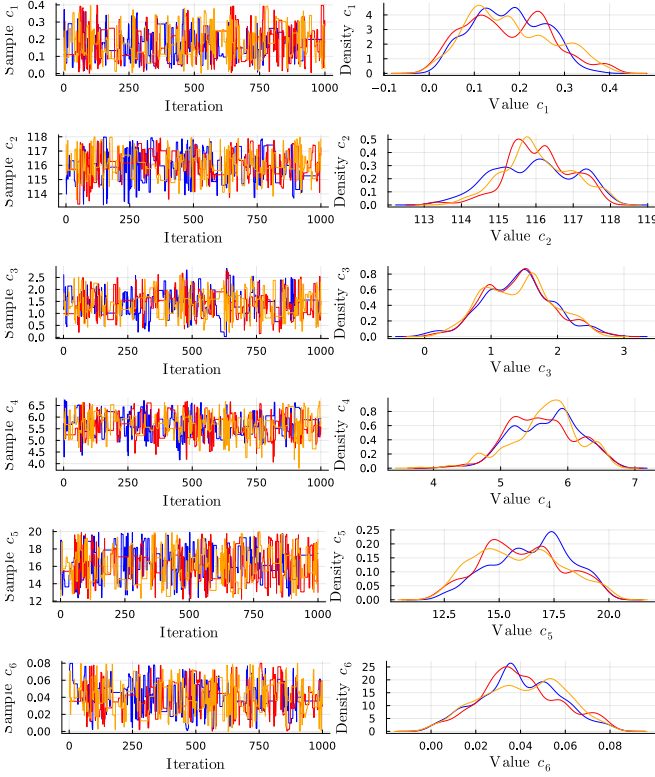


Fig. 3. Trace plots (left) and posterior density (right) of the estimated parameters c_1 - c_6 using three chains from the NUTS algorithm.

TABLE I
MODEL CONSTANTS.

Name	Description	Value
R	Rotor radius	120.998 m
A_r	Rotor area	πR^2 m ²
η	Generator efficiency	0.93
μ	Mechanical losses	0.05
ρ	Air density	1.225 kg m ⁻³

shows some standard literature values for the C_p model parameters based on [30], [35], which, however, do not accurately reflect the behaviour of the wind turbine studied here. Therefore, Bayesian inference with the No-U-Turn Sampler (NUTS) Hamiltonian Monte Carlo algorithm was employed using the Julia package TURING.JL to estimate the posterior distributions of these parameters (see Fig. 3). The estimation was based on the input data from the wind speed at the rotor v_r , and output data from rotor speed w_r , generator torque T_g and collective pitch angle β . A Gaussian approximation of the resulting posterior distributions is needed (see Table II). Controller gains were estimated via Levenberg-Marquardt least squares (see Table III) rather than treated probabilistically, as the setup involves open-loop detection where control signals are inputs of the filter. Consequently, the controller is not included in the filter model, and its gains are not required for filter design. The estimation of these gains was performed primarily for operating point computations and to support consistency checks.

TABLE II
 C_p PARAMETERS.

Name	Ref. value	μ	σ
c_1	0.5176	0.1766	0.0887
c_2	116	116.0236	0.9712
c_3	0.4	1.4138	0.5157
c_4	5	5.6603	0.4992
c_5	21	16.1613	1.8576
c_6	0.0068	0.0402	0.0178

TABLE III
CONTROLLER GAINS.

Name	Description	Value
K_{opt}	Optimal mode gain	3.2044×10^7
K_{p,T_g}	Proportional gain (torque)	30.2670
K_{i,T_g}	Integral gain (torque)	7.9968
$T_{g,rated}$	Rated generator torque	2.05×10^7
$K_{p,\beta}$	Proportional gain (pitch)	30.2670
$K_{i,\beta}$	Integral gain (pitch)	7.9968

V. IMPLEMENTATION AND RESULTS

Let $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{m \times n}$ represent the discrete-time system matrices of the wind turbine model, which depend on a set of stochastic parameters $\theta \in \mathbb{R}^l$ with mean values $\bar{\theta}$ and covariance matrix $\Sigma_\theta = \text{diag}(\sigma_{\theta_1}^2, \dots, \sigma_{\theta_l}^2)$, where $\theta_1, \dots, \theta_{l-1}$ are physical parameters ($c_1, \dots, c_6$) and θ_l represents the changes in operating points induced by changes in the operating point of the wind speed at the rotor, with gradients $d\omega_r/dv_r$, $d\omega_{r,err}/dv_r$, and $d\beta/dv_r$.

Hence, the total gradients of the system matrices with respect to all uncertain parameters θ_i ($i = 1, \dots, l$) are computed as

$$\nabla_{\theta_i} A = \frac{\partial A}{\partial \theta_i} \Big|_{\bar{\theta}, \omega_r^{\text{op}}, \beta^{\text{op}}}, \quad i = 1, \dots, l-1, \quad (40)$$

$$\nabla_{\theta_l} A = \left(\frac{\partial A}{\partial v_r} + \frac{\partial A}{\partial \omega_r} \frac{d\omega_r}{dv_r} + \frac{\partial A}{\partial \omega_{r,err}} \frac{d\omega_{r,err}}{dv_r} + \frac{\partial A}{\partial \beta} \frac{d\beta}{dv_r} \right) \Big|_{\bar{\theta}, \omega_r^{\text{op}}, \beta^{\text{op}}}, \quad (41)$$

$$\nabla_{\theta_i} B = \frac{\partial B}{\partial \theta_i} \Big|_{\bar{\theta}, \omega_r^{\text{op}}, \beta^{\text{op}}}, \quad i = 1, \dots, l-1, \quad (42)$$

$$\nabla_{\theta_l} B = \left(\frac{\partial B}{\partial v_r} + \frac{\partial B}{\partial \omega_r} \frac{d\omega_r}{dv_r} + \frac{\partial B}{\partial \omega_{r,err}} \frac{d\omega_{r,err}}{dv_r} + \frac{\partial B}{\partial \beta} \frac{d\beta}{dv_r} \right) \Big|_{\bar{\theta}, \omega_r^{\text{op}}, \beta^{\text{op}}}. \quad (43)$$

Note that v_r^{op} is considered in $\bar{\theta}$ ($\bar{\theta}_l$, specifically).

The matrices containing the standard deviations of each entry due to each uncertain parameter are computed as

$$A_{sd}^{(i)} = \sqrt{(\nabla_{\theta_i} A \circ \nabla_{\theta_i} A) \sigma_{\theta_i}^2}, \quad (44)$$

$$B_{sd}^{(i)} = \sqrt{(\nabla_{\theta_i} B \circ \nabla_{\theta_i} B) \sigma_{\theta_i}^2}, \quad (45)$$

where $\circ$ denotes the element-wise (Hadamard) product. Hence, the matrices A_i , B_i in (17) are defined as $A_i = \{A_{sd}^{(1)}, \dots, A_{sd}^{(l)}\}$, $B_i = \{B_{sd}^{(1)}, \dots, B_{sd}^{(l)}\}$. As the standard deviations are captured by A_i and B_i , the standard deviations σ_i of the stochastic process $p(k)$ will be defined as ones.

The reference model-based approach used in this paper is especially suited for addressing linearization errors, which arise because the model is only accurate near a specific operating point [21]. As the linearized model changes slightly with variations in wind speed v_r , we aim to ensure robustness against changes of up to 1 m/s. In contrast, uncertainty in the power coefficient model parameters $c_1, \dots, c_6$ stems from the use of a simplified model and parameter estimation. This uncertainty is time-invariant and therefore not considered in the LMI-based reference model filter design. However, it is taken into account in the residual evaluation stage via the residual covariance matrix $\Sigma_r(k)$ derived in (28). In summary, the filter design employs $A_i = A_{sd}^{(l)}$, $B_i = B_{sd}^{(l)}$, while the residual covariance derivation utilizes the full sets $A_i = \{A_{sd}^{(1)}, \dots, A_{sd}^{(l)}\}$, $B_i = \{B_{sd}^{(1)}, \dots, B_{sd}^{(l)}\}$.

Since the uncertain parameters influence only the aerodynamic model, only the first residual, r_{ω_r} , will exhibit non-zero bounds due to $\Sigma_r(k)$. The other two residuals, r_{T_g} and r_β , have confidence bounds determined solely by the assumed measurement noise, which is generally insufficient to capture transient spikes and other higher-order effects arising from unmodelled dynamics. Hence, relying on the raw $\Sigma_r(k)$ could allow sudden residual spikes to trigger crossings of the GLR decision threshold. To mitigate this, a small component corresponding to ≈ 2 times the nominal empirical residual variance of r_{T_g} and r_β was added to $\Sigma_r(k)$.

Due to this regularization term, using $Q_r = \Sigma_r^{-1}$ in (23) would yield misleading weights. Hence, Q_r is chosen from empirical sample variances of the reference residuals. Let $\hat{\sigma}_{r,i}^2 = \text{var}\{r_i(k)\}$ denote the sample variance of the i -th residual channel computed over a nominal (no-fault) segment of data. We form a diagonal weighting that equalizes the expected contribution of each channel in the bound:

$$Q_r = \text{diag}\left(\frac{\hat{\sigma}_{r,3}^2}{\hat{\sigma}_{r,1}^2}, \frac{\hat{\sigma}_{r,3}^2}{\hat{\sigma}_{r,2}^2}, 1\right). \quad (46)$$

Here the third component is taken as the reference scale ($q_{33} = 1$), so channels with larger empirical variance receive a proportionally smaller weight, and vice-versa.

The proposed scheme was validated through numerical experiments under the three attack scenarios mentioned in the end of Section IV. The robust residual generator was designed using the weighted LMI-based approach mentioned in Section III. The solution of the LMIs and Riccati equations is performed offline for each operating point. The optimization parameters were set to $\omega_1, \omega_2, \gamma_1, \gamma_2 = 1$, which resulted in $\alpha_1^2 = 1.7778$, $\alpha_2^2 = 0.8887$ in (24). For each scenario, the residuals generated by both the reference $\mathcal{H}_i/\mathcal{H}_\infty$ filter and the robust $\mathcal{H}_\infty$ MMP filter are shown in Figs. 4, 5, 6, along with their corresponding confidence bounds derived from the residual covariance due to parametric uncertainty $\Sigma_r(k)$ plus the regularization term.

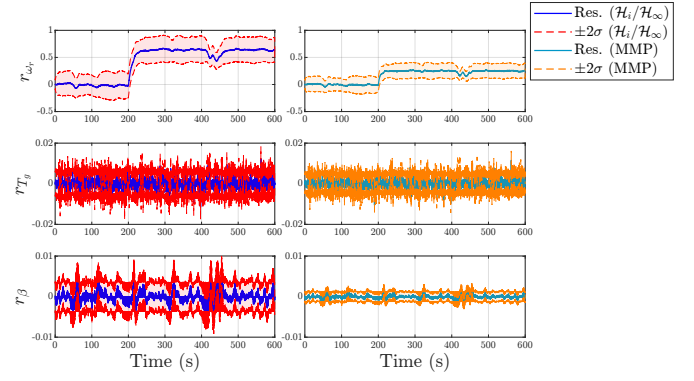


Fig. 4. Comparison between the $\mathcal{H}_i/\mathcal{H}_\infty$ residuals and the $\mathcal{H}_\infty$ MMP residuals for Test 1, shown together with 2σ confidence bounds.

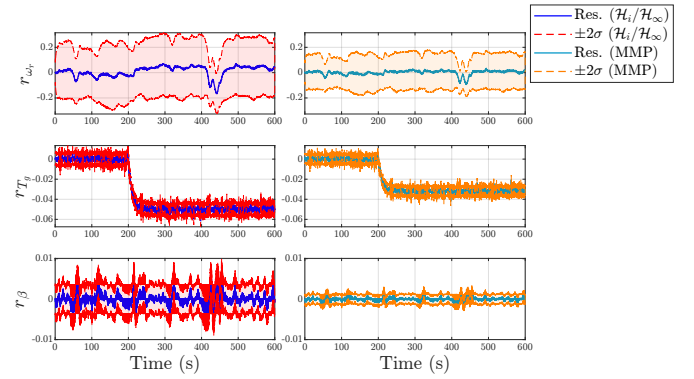


Fig. 5. Comparison between the $\mathcal{H}_i/\mathcal{H}_\infty$ residuals and the $\mathcal{H}_\infty$ MMP residuals for Test 2, shown together with 2σ confidence bounds.

Fig. 4 shows the residual behaviour for Test 1, where the bias injection attack is applied at $t = 200$ s, inducing a steady-state bias of $1 \cdot \omega_r^{\text{op}}$ into the measurement of ω_r . The controller compensates the false rotor speed increase by adjusting the pitch angle, which in turn causes the reduction of the actual rotor speed. This triggers the first reference residual, while the robust residuals show a reduced sensitivity, alongside a general decrease in residual variability.

Fig. 5 shows the residual behaviour for Test 2, where the attack is applied to the reference torque $T_{g,\text{ref}}$, inducing a steady-state bias of $-0.1 \cdot T_g^{\text{op}}$. To compensate and maintain the rotor speed operating point, the control system responds with a slight increase in pitch actuation, since pitch control has a significantly greater influence on rotor speed than torque control. As a result, the actual rotor speed slightly decreases, while the actual generator torque increases. The affected residuals are therefore r_{ω_r} and r_{T_g} .

Fig. 6 shows the residual behaviour for Test 3, where a bias injection attack is applied to the pitch reference signal β_{ref} with magnitude $0.1 \cdot \beta^{\text{op}}$. The system rapidly compensates for the induced bias to prevent a drop in rotor speed, resulting in a decrease in the actual pitch angle and a corresponding increase in the actual rotor speed. These deviations trigger responses in the residuals r_{ω_r} and r_β .

As shown in Fig. 4, Fig. 5, and Fig. 6, the uncertainty bounds plotted for the first residual r_{ω_r} do not always exhibit ideal behavior. For instance, during the larger perturbation

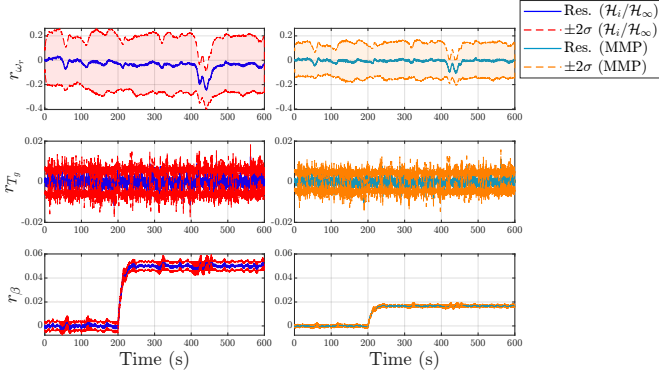


Fig. 6. Comparison between the $\mathcal{H}_i/\mathcal{H}_\infty$ residuals and the $\mathcal{H}_\infty$ MMP residuals for Test 3, shown together with 2σ confidence bounds.

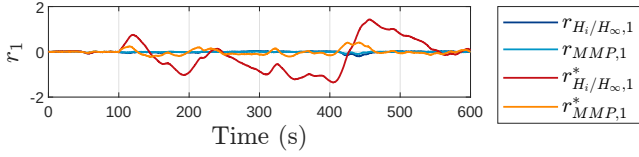


Fig. 7. Comparison between the first reference and robust residual in normal conditions, and the response with a change in the wind speed operating point from 14 m/s to 10 m/s at $t = 100$ s.

observed between 400 s and 500 s (not caused by the attack), one would expect the uncertainty bounds to widen accordingly. However, this does not occur. This discrepancy arises because the perturbation is not due to uncertainties in the C_p parameters or variations in wind speed, but rather to unmodelled dynamics or other forms of model mismatch that are not captured by the computed covariance. Despite this limitation, the resulting bounds represent the best achievable performance under the given modelling assumptions.

To more clearly illustrate the response of the MMP residuals to a considerable change in operating conditions, the operating point of the wind was changed from 14 m/s to 10 m/s at $t = 100$ s, as shown in Fig. 7. In the figure, the dark and light red curves represent the first $\mathcal{H}_i/\mathcal{H}_\infty$ reference residual and the $\mathcal{H}_\infty$ MMP robust residual for the perturbed system, respectively. For comparison, the dark and light blue curves show the same r_{w_r} residuals from Fig. 6. It is clear from the results that the robust residual is significantly less sensitive to the sudden change in operating conditions than the reference residual.

As explained in Section III, selecting thresholds based solely on the theoretical χ^2 distribution under $\mathcal{H}_0$ lead to overly optimistic false-alarm rates. Therefore, the threshold is chosen as the $(1 - \alpha)$ -quantile of the empirical distribution of $J(k)$ under attack-free operation to achieve a desired false-alarm probability α . For $\alpha = 0.5\%$, this yields $J_{th} = 152.96$ for the $\mathcal{H}_i/\mathcal{H}_\infty$ filter and $J_{th} = 85.54$ for the robust filter.

Robustness and sensitivity are evaluated under identical calibration conditions, with results summarized in Table IV. Under $\mathcal{H}_0$, the 99th percentile of $J(k)$ is lower for the robust filter ($q_{0.99} = 79.88$) than for the $\mathcal{H}_i/\mathcal{H}_\infty$ filter ($q_{0.99} = 126.55$), indicating reduced sensitivity to uncertainty. Under $\mathcal{H}_1$, both methods achieve a detection probability of 1 within a 10 s

TABLE IV
DETECTION PERFORMANCE ($\alpha = 0.5\%$).

Method	$q_{0.99}(\mathcal{H}_0)$	Delays [s]		
		T1	T2	T3
$\mathcal{H}_i/\mathcal{H}_\infty$	126.55	7.00	1.74	1.23
Robust	79.88	3.60	1.64	1.00

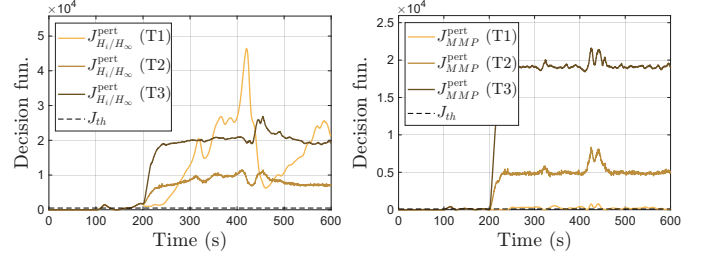


Fig. 8. GLR decision functions under operating-point perturbation (14 m/s $\rightarrow$ 10 m/s at $t = 100$ s) for $\mathcal{H}_i/\mathcal{H}_\infty$ (left) and robust filter (right).

window after the attack onset for all tested scenarios, while the robust filter yields equal or smaller detection delays (e.g., 3.6 s vs. 7 s in Test 1).

Regarding computational cost, the online implementation reduces to a linear observer update and the evaluation of the GLR test statistic. In the MATLAB implementation, the average execution time per iteration for the robust filter and GLR test statistic is 1.6×10^{-5} s and 4.16×10^{-4} s, respectively, which is well below the sampling period of 0.01 s.

Fig. 8 complements Fig. 7 by comparing the GLR test statistics of the reference and robust approaches under the change in operating conditions from 14 m/s to 10 m/s at $t = 100$ s. Although the robust residual generator is designed to handle variations of up to 1 m/s, a larger change of 4 m/s highlights more clearly its performance, resulting in a noticeable reduction in variability compared to the reference filter.

VI. CONCLUSION

This paper presents a reference model-based anomaly detection scheme for wind turbine systems subject to stochastic parameter uncertainties, with a particular focus on detecting false data injection cyber-attacks. The approach employs an $\mathcal{H}_i/\mathcal{H}_\infty$ reference filter, which offers an optimal balance between fault sensitivity and disturbance rejection, to design a robust filter through an $\mathcal{H}_\infty$ model matching framework. This filter tracks the reference residuals while accounting for parametric uncertainty. A closed-form expression for the residual covariance is derived from the statistical properties of the residual generator dynamics, providing a rigorous alternative to the common practice of inflating process noise covariances. Furthermore, the robust filter design is modified by weighting the influence of each residual component in the derivation of the residual variance bound to regulate conservativeness. Experimental results show a substantial reduction in residual sensitivity to uncertainty. The method's effectiveness strongly depends on accurate modelling of system uncertainties; other-

wise, it can become conservative. The findings highlight the potential of the proposed scheme for wind turbine monitoring and indicate several directions for improvement. One key direction involves developing a more rigorous approach to account for the residual variability caused by unmodelled dynamics, instead of adding a regularization term to the previously derived residual covariance. Future work will focus on this matter and the experimental validation of the method with operational data from real wind turbines.

ACKNOWLEDGMENT

We thank Emerson for providing data from a 15 MW wind turbine GH Bladed model.

APPENDIX

WEIGHTED REFERENCE MODEL-BASED LMI DESIGN FOR STOCHASTICALLY UNCERTAIN SYSTEMS

The LMIs related to (24) remain the same as in [21] (Section 8.5), whereas the LMI formulation for (25) in [21] is modified by introducing a weighting matrix $Q_r > 0$ into the residual covariance norm. Let us revise the whole procedure by first defining

$$V(k) = \mathcal{E} \left(x_r^T(k) \bar{P} x_r(k) \right), \quad (47)$$

where $\bar{P} > 0$, and decomposing it into mean and deviation,

$$V(k) = \text{trace}(\bar{P} E_{x_r}(k)) + \bar{x}_r^T(k) \bar{P} \bar{x}_r(k), \quad (48)$$

where $E_{x_r}(k) = \mathcal{E} (e_{x_r}(k) e_{x_r}^T(k))$ is the augmented state error covariance, with $e_{x_r}(k) = x_r(k) - \bar{x}_r(k)$.

The augmented state $x_r(k) = \begin{bmatrix} x(k) \\ x(k) - \hat{x}(k) \end{bmatrix}$ evolves as

$$x_r(k+1) = A_r x_r(k) + B_r u(k) + E_r d(k) + E_{r,f} f(k), \quad (49)$$

with $A_r = A_{r,0} + \sum_{i=1}^l A_{r,i} p_i(k)$, $B_r = B_{r,0} + \sum_{i=1}^l B_{r,i} p_i(k)$, and $E_r = E_{r,0} + \sum_{i=1}^l E_{r,i} p_i(k)$, where

$$\begin{aligned} A_{r,0} &= \begin{bmatrix} A & 0 \\ 0 & A - LC \end{bmatrix}, \quad A_{r,i} = \begin{bmatrix} A_i & 0 \\ A_i - LC_i & 0 \end{bmatrix}, \\ B_{r,0} &= \begin{bmatrix} B \\ 0 \end{bmatrix}, \quad B_{r,i} = \begin{bmatrix} B_i \\ B_i - LD_i \end{bmatrix}, \quad E_{r,0} = \begin{bmatrix} E_d \\ E_d - LF_d \end{bmatrix}, \\ E_{r,i} &= \begin{bmatrix} E_i \\ E_i - LF_i \end{bmatrix}, \quad E_{r,f} = \begin{bmatrix} E_f \\ E_f - LF_f \end{bmatrix}. \end{aligned}$$

The mean $\bar{x}_r(k+1)$ and the deviation from the mean $e_{x_r}(k+1) = x_r(k+1) - \bar{x}_r(k+1)$ are thus expressed as

$$\bar{x}_r(k+1) = A_{r,0} \bar{x}_r(k) + B_{r,0} u(k) + E_{r,0} d(k) + E_{r,f} f(k), \quad (50)$$

$$\begin{aligned} e_{x_r}(k+1) &= A_{r,0} e_{x_r}(k) + \sum_{i=1}^l A_{r,i} p_i(k) x_r(k) \\ &\quad + \sum_{i=1}^l B_{r,i} p_i(k) u(k) + \sum_{i=1}^l E_{r,i} p_i(k) d(k). \end{aligned} \quad (51)$$

Expanding $E_{x_r}(k+1)$ into the resulting terms and applying the expectations yields

$$\begin{aligned} E_{x_r}(k+1) &= A_{r,0} E_{x_r}(k) A_{r,0}^T \\ &\quad + \sum_i \sigma_i^2 A_{r,i} (\bar{x}_r(k) \bar{x}_r^T(k) + E_{x_r}(k)) A_{r,i}^T \\ &\quad + \sum_i \sigma_i^2 [B_{r,i} u(k) u(k)^T B_{r,i}^T + E_{r,i} d(k) d(k)^T E_{r,i}^T] \\ &\quad + \sum_i \sigma_i^2 [A_{r,i} \bar{x}_r(k) u(k)^T B_{r,i}^T + B_{r,i} u(k) \bar{x}_r(k)^T A_{r,i}^T] \\ &\quad + \sum_i \sigma_i^2 [A_{r,i} \bar{x}_r(k) d(k)^T E_{r,i}^T + E_{r,i} d(k) \bar{x}_r(k)^T A_{r,i}^T] \\ &\quad + \sum_i \sigma_i^2 [B_{r,i} u(k) d(k)^T E_{r,i}^T + E_{r,i} d(k) u(k)^T B_{r,i}^T]. \end{aligned} \quad (52)$$

This allows the definition of $V(k+1)$ as

$$V(k+1) = \text{trace}(M_A E_{x_r}(k)) + \begin{bmatrix} \bar{x}_r(k) \\ u(k) \\ d(k) \\ f(k) \end{bmatrix}^T M_1 \begin{bmatrix} \bar{x}_r(k) \\ u(k) \\ d(k) \\ f(k) \end{bmatrix}, \quad (53)$$

where

$$M_A = A_{r,0}^T \bar{P} A_{r,0} + \sum_{i=1}^l \sigma_i^2 A_{r,i}^T \bar{P} A_{r,i}, \quad (54)$$

$$\begin{aligned} M_1 &= \begin{bmatrix} A_{r,0}^T \\ B_{r,0}^T \\ E_{r,0}^T \\ E_{r,f}^T \end{bmatrix} \bar{P} \begin{bmatrix} A_{r,0} & B_{r,0} & E_{r,0} & E_{r,f} \end{bmatrix} \\ &\quad + \sum_{i=1}^l \sigma_i^2 \begin{bmatrix} A_{r,i}^T \\ B_{r,i}^T \\ E_{r,i}^T \\ 0 \end{bmatrix} \bar{P} \begin{bmatrix} A_{r,i} & B_{r,i} & E_{r,i} & 0 \end{bmatrix}. \end{aligned} \quad (55)$$

Assuming

$$M_1 < \begin{bmatrix} \bar{P} & 0 & 0 & 0 \\ 0 & I & 0 & 0 \\ 0 & 0 & I & 0 \\ 0 & 0 & 0 & I \end{bmatrix} \Rightarrow M_A \leq \bar{P}, \quad (56)$$

yields

$$V(k+1) < V(k) + d^T(k) d(k) + u^T(k) u(k) + f^T(k) f(k), \quad (57)$$

which leads to

$$V(k) < \sum_{j=0}^{k-1} [d^T(j) d(j) + u^T(j) u(j) + f^T(j) f(j)]. \quad (58)$$

Now, consider the weighted residual covariance norm

$$\sigma_r^2(k) = \mathcal{E} ((r(k) - \bar{r}(k))^T Q_r (r(k) - \bar{r}(k))). \quad (59)$$

The residual is given by:

$$r(k) = C_r x_r(k) + D_r u(k) + F_r d(k) + F_{r,f} f(k), \quad (60)$$

with $C_r = C_{r,0} + \sum_{i=1}^l C_{r,i} p_i(k)$, $D_r = \sum_{i=1}^l D_{r,i} p_i(k)$, and $F_r = F_{r,0} + \sum_{i=1}^l F_{r,i} p_i(k)$, where

$$\begin{aligned} C_{r,0} &= [0 \quad VC], \quad C_{r,i} = [VC_i \quad 0], \\ D_{r,i} &= VD_i, \quad F_{r,0} = VF_d, \quad F_{r,i} = VF_i, \quad F_{r,f} = VF_f. \end{aligned}$$

The mean of the residual is

$$\bar{r}(k) = \mathcal{E}[r(k)] = C_{r,0}\bar{x}_r(k) + F_{r,0}d(k) + F_{r,f}f(k). \quad (61)$$

Expanding $\sigma_r^2(k)$ and applying the expectations to each term yields

$$\begin{aligned} \sigma_r^2(k) = & \text{trace}(Q_r C_{r,0} E_{x_r}(k) C_{r,0}^T) \\ & + \sum_{i=1}^l \sigma_i^2 \text{trace}(Q_r C_{r,i} E_{x_r}(k) C_{r,i}^T) \\ & + \sum_{i=1}^l \sigma_i^2 \bar{x}_r(k)^T C_{r,i}^T Q_r C_{r,i} \bar{x}_r(k) \\ & + \sum_{i=1}^l \sigma_i^2 u(k)^T D_{r,i}^T Q_r D_{r,i} u(k) \\ & + \sum_{i=1}^l \sigma_i^2 d(k)^T F_{r,i}^T Q_r F_{r,i} d(k) \\ & + \sum_{i=1}^l \sigma_i^2 [D_{r,i} u(k) d(k)^T F_{r,i}^T + F_{r,i} d(k) u(k)^T D_{r,i}^T]. \end{aligned} \quad (62)$$

The mixed terms involving $\hat{x}_r(k)$ with $u(k)$ and $d(k)$ are omitted to obtain an LMI condition that is independent of the unknown mean trajectory $\hat{x}_r(k)$. The weighted residual covariance norm can be rewritten as

$$\begin{aligned} \sigma_r^2(k) = & \text{trace}(Q_r M_C E_{x_r}(k)) + \bar{x}_r^T(k) Q_r M_2 \bar{x}_r(k) \\ & + \begin{bmatrix} u(k) \\ d(k) \\ f(k) \end{bmatrix}^T Q_r M_3 \begin{bmatrix} u(k) \\ d(k) \\ f(k) \end{bmatrix}, \end{aligned} \quad (63)$$

with:

$$\begin{aligned} M_C &= C_{r,0}^T C_{r,0} + \sum_{i=1}^l \sigma_i^2 C_{r,i}^T C_{r,i}, \\ M_2 &= \sum_{i=1}^l \sigma_i^2 C_{r,i}^T C_{r,i} \leq M_C, \\ M_3 &= \sum_{i=1}^l \sigma_i^2 \begin{bmatrix} D_{r,i}^T \\ F_{r,i}^T \\ 0 \end{bmatrix} [D_{r,i} \quad F_{r,i} \quad 0]. \end{aligned}$$

Imposing the condition

$$\begin{aligned} & \text{trace}(Q_r M_C E_{x_r}(k)) + \bar{x}_r^T(k) Q_r M_2 \bar{x}_r(k) \\ & \leq \gamma_1^2 (\text{trace}(\bar{P} E_{x_r}(k)) + \bar{x}_r^T(k) \bar{P} \bar{x}_r(k)), \end{aligned} \quad (64)$$

which holds when

$$Q_r M_C \leq \gamma_1^2 \bar{P}, \quad (65)$$

and also

$$Q_r M_3 \leq \gamma_2^2 I, \quad (66)$$

leads to (25)

$$\begin{aligned} \sigma_r^2(k) & < \gamma_1^2 \sum_{j=0}^{k-1} [d^T(j) d(j) + u^T(j) u(j) + f^T(j) f(j)] \\ & + \gamma_2^2 [d^T(k) d(k) + u^T(k) u(k) + f^T(k) f(k)]. \end{aligned}$$

The previous LMIs (56), (65) and (66) are reformulated by redefining M_1 , M_C and M_3 as

$$\begin{aligned} N_0 &= [A_{r,0} \quad B_{r,0} \quad E_{r,0} \quad E_{r,f}], \\ N_i &= [A_{r,i} \quad B_{r,i} \quad E_{r,i} \quad 0], \quad i = 1, \dots, l, \end{aligned}$$

$$M_1 = \begin{bmatrix} N_0^T \\ N_1^T \\ \vdots \\ N_l^T \end{bmatrix}^T \begin{bmatrix} \bar{P} & 0 & \cdots & 0 \\ 0 & \sigma_1^2 \bar{P} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \sigma_l^2 \bar{P} \end{bmatrix} \begin{bmatrix} N_0 \\ N_1 \\ \vdots \\ N_l \end{bmatrix}, \quad (67)$$

$$M_C = \begin{bmatrix} C_{r,0}^T \\ C_{r,1}^T \\ \vdots \\ C_{r,l}^T \end{bmatrix}^T \begin{bmatrix} Q_r & 0 & \cdots & 0 \\ 0 & \sigma_1^2 Q_r & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \sigma_l^2 Q_r \end{bmatrix} \begin{bmatrix} C_{r,0} \\ C_{r,1} \\ \vdots \\ C_{r,l} \end{bmatrix}, \quad (68)$$

$$T_{r,i} = [V D_i \quad V F_i \quad 0], \quad i = 1, \dots, l,$$

$$M_3 = \begin{bmatrix} T_{r,1}^T \\ \vdots \\ T_{r,l}^T \end{bmatrix}^T \begin{bmatrix} \sigma_1^2 Q_r & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_l^2 Q_r \end{bmatrix} \begin{bmatrix} T_{r,1} \\ \vdots \\ T_{r,l} \end{bmatrix}. \quad (69)$$

Using the Schur complement, LMI (56) becomes

$$\begin{bmatrix} -\bar{P} & N_0^T \bar{P} & N_1^T \bar{P} & \cdots & N_l^T \bar{P} \\ \bar{P} N_0 & -\bar{P} & 0 & \cdots & 0 \\ \bar{P} N_1 & 0 & -\sigma_1^{-2} \bar{P} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \bar{P} N_l & 0 & \cdots & 0 & -\sigma_l^{-2} \bar{P} \end{bmatrix} < 0, \quad (70)$$

where

$$\tilde{P} = \begin{bmatrix} \bar{P} & 0 & 0 & 0 \\ 0 & I & 0 & 0 \\ 0 & 0 & I & 0 \\ 0 & 0 & 0 & I \end{bmatrix}.$$

LMI (65) becomes

$$\begin{bmatrix} -\gamma_1^2 \bar{P} & C_{r,0}^T & C_{r,1}^T & \cdots & C_{r,l}^T \\ C_{r,0} & -Q_r^{-1} & 0 & \cdots & 0 \\ C_{r,1} & 0 & -\sigma_1^{-2} Q_r^{-1} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ C_{r,l} & 0 & \cdots & 0 & -\sigma_l^{-2} Q_r^{-1} \end{bmatrix} \leq 0, \quad (71)$$

LMI (66) becomes

$$\begin{bmatrix} -\gamma_2^2 I & T_{r,1}^T & \cdots & T_{r,l}^T \\ T_{r,1} & -\sigma_1^{-2} Q_r^{-1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ T_{r,l} & \cdots & 0 & -\sigma_l^{-2} Q_r^{-1} \end{bmatrix} \leq 0. \quad (72)$$

The optimization problem becomes

$$\min_{L,V} (w_1 \alpha_1^2 + w_2 \alpha_2^2), \quad (73)$$

subject to

- LMIs (70)–(72) using

$$P = \begin{bmatrix} P_1 & 0 \\ 0 & P_2 \end{bmatrix}, \quad \bar{P} = \begin{bmatrix} P_3 & 0 \\ 0 & P_1 \end{bmatrix},$$

with

$$\bar{P}N_0 = \begin{bmatrix} P_3A & 0 & P_3B \\ 0 & P_1A - YC & 0 \\ P_3E_d & P_3E_f \\ P_1E_d - YF_d & P_1E_f - YF_f \end{bmatrix}, \quad (74)$$

$$\bar{P}N_i = \begin{bmatrix} P_3A_i & 0 & P_3B_i \\ 0 & P_1A_i - YC_i & P_1B_i - YD_i \\ P_3E_i & 0 \\ P_1E_i - YF_i & 0 \end{bmatrix}. \quad (75)$$

- Residual filter gain: $L = P_1^{-1}Y$.
- Residual variance LMIs (70)–(72) with Q_r .

REFERENCES

- [1] S. G. Freeman, M. A. Kress-Weitenhagen, J. P. Gentle, M. J. Culler, M. M. Egan, and R. V. Stolworthy, "Attack surface of wind energy technologies in the united states," Idaho National Laboratory (INL), Idaho Falls, ID (United States), Tech. Rep., 2024.
- [2] J. A. Guerrero-Saade and M. Van Amerongen, "Acidrain— a modem wiper rains down on europe," *Sentinel Labs*, vol. 31, 2022.
- [3] M. Döhler, L. Mevel, and Q. Zhang, "Fault detection, isolation and quantification from gaussian residuals with application to structural damage diagnosis," *Annual Reviews in Control*, vol. 42, pp. 244–256, 2016.
- [4] S. Simani, P. Castaldi *et al.*, "Fault diagnosis techniques for a wind turbine system," *Fault Detection, Diagnosis and Prognosis*, pp. 1–20, 2019.
- [5] J. Liang, K. Zhang, A. Al-Durra, S. Mueen, and D. Zhou, "A state-of-the-art review on wind power converter fault diagnosis," *Energy Reports*, vol. 8, pp. 5341–5369, 2022.
- [6] M. Ghiasi and M. Fotuhi-Firuzabad, "Resilience enhancement of smart power systems against false data injection attacks using adaptive intrusion detection mechanisms," *IEEE Transactions on Reliability*, 2026.
- [7] S. X. Ding, *Advanced methods for fault diagnosis and fault-tolerant control*. Springer, 2021.
- [8] M. Hansen, *Aerodynamics of wind turbines*. Routledge, 2015.
- [9] E.-J. Perez-Perez, V. Puig, F.-R. López-Estrada, G. Valencia-Palomo, I. Santos-Ruiz, and S. E. Samada, "Fault detection and isolation in wind turbines based on neuro-fuzzy qlpv zonotopic observers," *Mechanical Systems and Signal Processing*, vol. 191, p. 110183, 2023.
- [10] E.-J. Pérez-Pérez, V. Puig, F.-R. López-Estrada, G. Valencia-Palomo, I. Santos-Ruiz, and G. Osorio-Gordillo, "Robust fault diagnosis of wind turbines based on manifolds and zonotopic observers," *Expert Systems with Applications*, vol. 235, p. 121095, 2024.
- [11] X. Ding and P. M. Frank, "Fault detection via factorization approach," *Systems & control letters*, vol. 14, no. 5, pp. 431–436, 1990.
- [12] M. Hou and R. J. Patton, "Input observability and input reconstruction," *Automatica*, vol. 34, no. 6, pp. 789–794, 1998.
- [13] Q. Zhang and L. Zhang, "State estimation for stochastic time varying systems with disturbance rejection**l. zhang acknowledges the financial support partly by the national nature science foundation of china (11701040) and innovation foundation of bupt for youth (500417024)." *IFAC-PapersOnLine*, vol. 51, no. 15, pp. 55–59, 2018, 18th IFAC Symposium on System Identification SYSID 2018.
- [14] M. Zhong, S. X. Ding, J. Lam, and H. Wang, "An lmi approach to design robust fault detection filter for uncertain lti systems," *Automatica*, vol. 39, no. 3, pp. 543–550, 2003.
- [15] D. Sauter and F. Hamelin, "Frequency-domain optimization for robust fault detection and isolation in dynamic systems," *IEEE Transactions on automatic control*, vol. 44, no. 4, pp. 878–882, 2002.
- [16] J. Liu, J. L. Wang, and G.-H. Yang, "An lmi approach to minimum sensitivity analysis with application to fault detection," *Automatica*, vol. 41, no. 11, pp. 1995–2004, 2005.
- [17] J. L. Wang, G.-H. Yang, and J. Liu, "An lmi approach to $\mathcal{H}_\infty$ index and mixed $\mathcal{H}_\infty/\mathcal{H}_2$ fault detection observer design," *Automatica*, vol. 43, no. 9, pp. 1656–1665, 2007.
- [18] S. X. Ding, T. Jeinsch, P. M. Frank, and E. L. Ding, "A unified approach to the optimization of fault detection systems," *International journal of adaptive control and signal processing*, vol. 14, no. 7, pp. 725–745, 2000.
- [19] P. Zhang, S. X. Ding, G. Wang, and D. Zhou, "Fault detection of linear discrete-time periodic systems," *IEEE Transactions on Automatic Control*, vol. 50, no. 2, pp. 239–244, 2005.
- [20] X. Li and K. Zhou, "A time domain approach to robust fault detection of linear time-varying systems," *Automatica*, vol. 45, no. 1, pp. 94–102, 2009.
- [21] S. X. Ding, *Model-based fault diagnosis techniques: design schemes, algorithms, and tools*. Springer Science & Business Media, 2008.
- [22] M. Zhong, S. X. Ding, B. Tang, P. Zhang, and T. Jeinsch, "An lmi approach to robust fault detection filter design for discrete-time systems with model uncertainty," in *Proceedings of the 40th IEEE Conference on Decision and Control (Cat. No. 01CH37228)*, vol. 4. IEEE, 2001, pp. 3613–3618.
- [23] E. Frisk and L. Nielsen, "Robust residual generation for diagnosis including a reference model for residual behavior," *Automatica*, vol. 42, no. 3, pp. 437–445, 2006.
- [24] E. Mazars, I. Jaimoukha, Z. Li, and A. C. Zolotas, "Fault detection and isolation filter design for systems subject to polytopic uncertainties," in *2007 Mediterranean Conference on Control & Automation*. IEEE, 2007, pp. 1–5.
- [25] M. Nazir, A. Q. Khan, G. Mustafa, and M. Abid, "Robust fault detection for wind turbines using reference model-based approach," *Journal of King Saud University-Engineering Sciences*, vol. 29, no. 3, pp. 244–252, 2017.
- [26] Á. M. Gómez, S. Hassani, S. Gres, and R. Wisniewski, "Reference model-based cyber-attack detection for stochastically uncertain wind turbine systems," in *2025 IEEE Industry Applications Society Annual Meeting (IAS)*. IEEE, 2025, pp. 1–8.
- [27] A. Teixeira, I. Shames, H. Sandberg, and K. H. Johansson, "A secure control framework for resource-limited adversaries," *Automatica*, vol. 51, pp. 135–148, 2015.
- [28] M. Basseville, I. V. Nikiforov *et al.*, *Detection of abrupt changes: theory and application*. prentice Hall Englewood Cliffs, 1993, vol. 104.
- [29] Á. M. Gómez, R. Wisniewski, and L. Navarro-Hilfiker, "Model-based detection of data-injection cyber-attacks on wind turbine controllers," in *International Conference on Control, Decision and Information Technologies*, 2024.
- [30] M. Carpintero-Renteria, D. Santos-Martin, A. Lent, and C. Ramos, "Wind turbine power coefficient models based on neural networks and polynomial fitting," *IET Renewable Power Generation*, vol. 14, no. 11, pp. 1841–1849, 2020.
- [31] T. Knudsen and T. Bak, "Simple model for describing and estimating wind turbine dynamic inflow," in *2013 American Control Conference*. IEEE, 2013, pp. 640–646.
- [32] M. Jelavić and N. Perić, "Wind turbine control for high turbulent winds," *Automatika: časopis za automatiku, mjerenje, elektroniku, računarstvo i komunikacije*, vol. 50, no. 3-4, pp. 135–151, 2009.
- [33] T. Knudsen, T. Bak, and M. Soltani, "Prediction models for wind speed at turbine locations in a wind farm," *Wind Energy*, vol. 14, no. 7, pp. 877–894, 2011.
- [34] E. Bossanyi, "Gh bladed user manual," *Garrad Hassan and Partners Ltd*, vol. 701, 2009.
- [35] S. Heier, *Grid integration of wind energy: onshore and offshore conversion systems*. John Wiley & Sons, 2014.